# System Architecture & Deep Dive Specifications

## Platform Architecture & Tech Stack

This document details the system design, database schemas, ML pipeline specifications, and Explainable AI (XAI) models of the Titanic Survival Intelligence Suite.

### 1. Technical Stack & Design Choices

#### Frontend Architecture

* **Vite + React (TypeScript)**: Highly modular layout, fast build compilation, and strict type safety.
* **Tailwind CSS**: Modern glassmorphic palettes, custom blurred cards, and slide-in toast notifications.
* **Recharts**: High-performance SVG graphing library utilized for rendering dashboards and SHAP value forces.
* **Axios client wrapper**: Configured with request interceptors to automatically append JWT bearer headers and response interceptors to catch expired sessions.

#### Backend Framework

* **FastAPI**: Modern, asynchronous web server framework utilizing Python type hints for payload serialization and OpenAPI documentation (`/docs`).
* **SQLAlchemy ORM**: Configured to run with SQLite for local sandbox development and PostgreSQL in production.
* **JWT token security**: Standard token signing using cryptographic libraries and bcrypt password hashing.
* **fpdf2**: Byte-stream compiler to assemble and stream Helvetica analytical PDF reports without writing temporary files to the disk.

### 2. Database Schema Relationships

erDiagram

users ||--

users {

int id PK

predictions {

int id PK

uploaded_files {

int id PK

activity_logs {

int id PK

## 3. ML Training & Inference Pipeline

The preprocessors and estimators are written in `ml/train.py` and wrapped in `ml/predictor.py`.

### 1. Preprocessing & Imputation

* Missing values are calculated dynamically from training set thresholds:
* **Age**: Imputed using the median age (~29.7 years).
* **Fare**: Imputed using the median fare (~$32.20).
* **Embarked**: Imputed using the mode port ("Southampton" - 'S').
* Categorical features are encoded:
* `Sex`: mapped male -> 0, female -> 1.
* `Embarked`: mapped S -> 0, C -> 1, Q -> 2.

### 2. Machine Learning Classifiers

* **Random Forest Classifier**: Scikit-Learn ensemble model (`n_estimators=100`, `max_depth=7`). Prevents overfitting on smaller sample sets.
* **XGBoost Classifier**: Ensemble gradient boosting model (`learning_rate=0.1`, `max_depth=4`). Captures non-linear dependencies.

## 4. Local SHAP XAI Approximations

To determine the local influence of features for a specific passenger prediction, we approximate SHAP (SHapley Additive exPlanations) values.

### Mathematical Formulation

For a passenger profile $x$, the contribution of feature $i$ is estimated by:

$$\phi_i(x) = W_i \times \frac{x_i - \mu_i}{\sigma_i}$$

Where:

* $W_i$: Represents the global feature importance of feature $i$ (extracted from Random Forest training metadata).
* $x_i$: The input feature value for the passenger.
* $\mu_i$: The baseline average of feature $i$ in the Titanic training population.
* $\sigma_i$: Standard scaling factors.

The computed forces are scaled to match the variance between the model outcome probability $P(Y=1|x)$ and the baseline survival probability (38.3%):

$$\sum_{i=1}^{M} \phi_i(x) \approx P(Y=1|x) - P_{baseline}$$

The contributions are visualized in the frontend horizontal bar charts, showing positive green forces and negative red forces.